

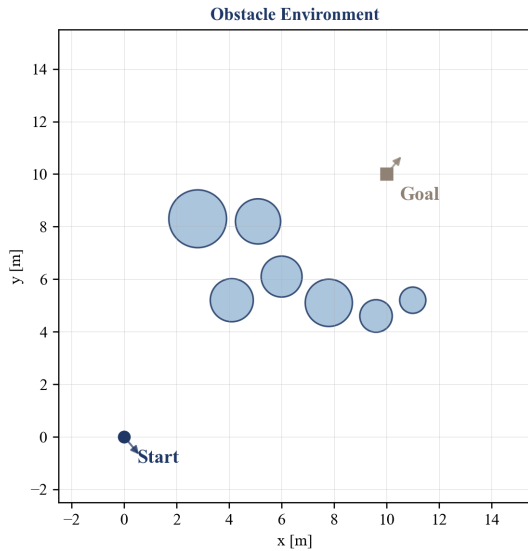
AER1516: Robot Motion Planning

Assignment #2 — RRT vs. RRT* Comparison

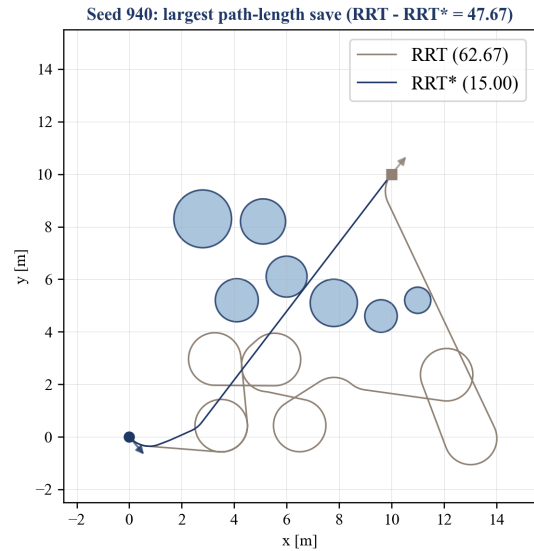
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1 Obstacle Environment and Example Paths

The environment is a $[-2, 15] \times [-2, 15]$ map with seven circular obstacles clustered across the mid-field. Because the obstacles sit between the start and goal, the Dubins vehicle must curve around them, creating opportunities where **RRT** returns the first feasible path (often with large detours) while **RRT*** continues rewiring to find shorter routes.



(a) Obstacle environment with start/goal poses



(b) Example: largest path-length improvement

2 Performance Comparison

Both planners were run for **1,000 random instances** with `max_iter = 10,000`, using seeds `0, 1, ..., 999`. The bar chart shows **RRT*** achieves a lower average path length than **RRT**, confirming the asymptotic optimality advantage.

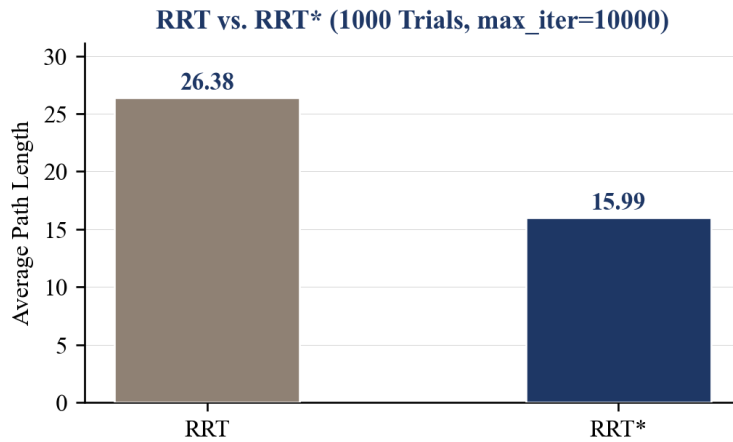


Figure 2: Average path length comparison over 1,000 trials